

LOKESH DUMPA

Software Engineer | Portfolio: Lokesh

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Summary

Experienced Software Engineer with 5 years of expertise in backend development, DevOps, and machine learning, achieving a 40% reduction in deployment time through Kubernetes and Jenkins automation. Led the design of scalable RESTful APIs and infrastructure provisioning using Python, AWS, and Terraform. Engineered CI/CD pipelines with GitLab CI/CD and GitHub Actions to support uninterrupted production releases. Developed secure, containerized microservices compliant with IAM policies and HIPAA standards. Applied Spark, Pandas, and PySpark for efficient data processing and feature engineering. Integrated comprehensive logging and monitoring solutions using CloudWatch, Grafana, and ELK Stack. Delivered end-to-end ML pipelines for real-time analytics and enhanced cloud resource usage.

Technical Skills

- **Programming Languages:** Python, Java, C, JavaScript, Shell Scripting, Go, Object-Oriented Programming
- **Web Development:** HTML, CSS, React
- **Backend Development:** Node.js
- **Cloud Platforms & Services:** AWS, Azure, Google Cloud Platform
- **Infrastructure & DevOps Tools:** Terraform, Docker, Kubernetes, Jenkins, Git, GitLab CI/CD, GitHub Actions, CloudFormation, GitOps, Helm, Linux, Bash, IAM policies
- **Monitoring, Logging & Security:** CloudWatch, CloudTrail, SonarQube, Datadog, Cloud Security, Grafana, Nagios, Elasticsearch
- **Databases:** SQL, PostgreSQL, MySQL, Amazon RDS, NoSQL
- **Data Visualization Tools:** Power BI, Tableau
- **Data Warehousing & ETL:** Redshift, Glue, Athena
- **Big Data & Distributed Computing:** PySpark, Databricks
- **Python Libraries:** Pandas, NumPy, Matplotlib, Seaborn
- **Machine Learning & AI:** Deep Learning, NLP, LLMs, Scikit-learn, TensorFlow, Keras, OpenCV, Transformers, Streamlit
- **Project Management Tools:** Confluence, JIRA
- **Development Methodologies:** Agile
- **Testing Frameworks:** TDD (Pytest, unittest)
- Data Structures & Algorithms
- System Design, Design Patterns
- Software Development Life Cycle

Professional Experience

University of South Florida

January 2024 – May 2025

Research Assistant

Technologies: Python, Shell, Apache Spark, Pandas, MLflow, AWS (EC2, S3, Lambda, IAM), Docker, Kubernetes, Terraform, Ansible, Jenkins, Git, CloudWatch

- Designed and streamlined machine learning pipelines by automating tasks and leveraging scalable cloud infrastructure using Python, AWS, and Docker.
- Led the deployment and CI/CD of ML models with Jenkins and Kubernetes, reducing deployment time by 40% and ensuring production stability.
- Established strong data acquisition and preprocessing systems using Apache Spark and Pandas, managing large-scale datasets to refine model accuracy.
- Partnered with research and IT teams to provision infrastructure-as-code through Terraform and Ansible, cutting manual setup time by 50% and boosting reproducibility.
- Executed end-to-end model evaluation and monitored performance using MLflow and custom dashboards to detect data drift and ensure model stability.

Tata Consultancy Services

October 2021 – August 2023

Software Engineer

Project Name: CoolApps

Technologies: Python, AWS (EC2, Lambda, RDS, EMR, CloudWatch, IAM), PostgreSQL, Terraform, GitLab CI/CD, GitHub Actions, Kubernetes, Helm, Liquibase, Pytest, SonarQube, Elastic Stack, REST APIs, ReactJS

- Architected and delivered scalable RESTful APIs using Python and AWS (Lambda, EC2, RDS) to support high-availability, low-latency backend services across many environments.
- Automated cloud infrastructure provisioning and CI/CD pipelines with Terraform and GitLab, accelerating deployment frequency by 40% while ensuring zero downtime.
- Optimized complex PostgreSQL queries and refactored backend logic, enhancing database performance and reducing query execution time.
- Integrated advanced monitoring solutions leveraging AWS CloudWatch and Python, leading to a 20% reduction in incident response time.

- Directed implementation of IAM-based security policies and containerized microservices deployment through GitHub Actions with Pytest and SonarQube, increasing release quality and security posture.

Tata Consultancy Services

August 2021 – January 2022

Software Engineer

Technologies: Python, AWS (EC2, IAM, Lambda), Docker, RESTful APIs, Machine Learning, Parquet, CSV, Multithreading, Confluence, Cloud-native architecture.

- Engineered a scalable full-stack medical analytics platform using Python, AWS EC2, and RESTful APIs to accelerate real-time disease diagnosis and improve clinical workflow efficiency.
- Created strong ML classification pipelines processing diverse medical records (CSV, Parquet) with modular Python code, improving model accuracy by 18% and ensuring smooth data ingestion.
- Architected backend services with Docker and secured cloud deployments via AWS IAM, ensuring compliance with HIPAA standards and promoting secure infrastructure operations.
- Improved API response time by 35% through multithreading and resource tuning on AWS infrastructure, sustaining high throughput under live inference workloads.
- Facilitated cross-team coordination through stakeholder sync-ups and architectural reviews, aligning model retraining pipelines and cloud SLAs with project milestones.

Cynnent

March 2020 – May 2021

DevOps Engineer

Technologies: Jenkins, GitHub Actions, AWS (EC2, S3, IAM, RDS), Terraform, Docker, Kubernetes, Prometheus, Grafana, ELK Stack, Ansible, Python, Agile, CI/CD, Infrastructure as Code, Cloud Security

- Engineered CI/CD pipelines using Jenkins and GitHub Actions, streamlining deployments and reducing release errors, enhancing overall software delivery efficiency.
- Designed cloud infrastructure on AWS EC2, S3, IAM, and RDS leveraging Terraform, accelerating setup times by 70% and ensuring environment consistency across development, staging, and production.
- Managed containerized applications using Docker and Kubernetes, improving scalability and operational performance, while enabling zero-downtime deployments for critical applications.
- Established comprehensive monitoring and alerting systems with Prometheus, Grafana, and ELK Stack, resulting in a 50% reduction in incident response time and improved system reliability.
- Developed automation scripts with Ansible and Python to streamline server configuration, enforce security policies, and support Agile DevOps practices, contributing to operational efficiency gains.

Projects

DevOps Intelligence Dashboard with Power BI

Designed and implemented a real-time DevOps analytics dashboard to monitor CI/CD performance and AWS cloud spend.

- Automated daily data extraction from GitHub Actions logs and AWS Cost Explorer APIs using Python scripts.
- Transformed JSON pipeline logs into structured metrics and CSV datasets for seamless Power BI ingestion.
- Visualized DevOps KPIs such as Build Success Rate, Deployment Frequency, and AWS daily spend via interactive Power BI dashboards.
- Engineered dynamic DAX measures and filters (repo, environment, build status) for actionable, drill-down insights.
- Deployed and documented the Power BI (.pbix) dashboard with live data for GitHub sharing and stakeholder reviews.

GenAIOps | Azure, Terraform, Kubernetes

Designed and deployed a production-grade AI microservices pipeline on Azure using Terraform and Kubernetes.

- Provisioned scalable Azure infrastructure (AKS, ACR, VNet) following Infrastructure-as-Code principles with Terraform.
- Containerized and deployed Python (Flask) and Java (Spring Boot) microservices to Azure Kubernetes Service for modular AI workflows.
- Set up CI/CD pipelines using GitHub Actions for automated Docker image build, push, and AKS deployment, ensuring rapid delivery.
- Configured secure service-to-service communication via Kubernetes DNS and secrets management (OpenAI API keys).
- Tested real-time sentiment generation by integrating AI-powered Flask services with Java APIs, validating end-to-end interactions on AKS.

Education

Master of Science in Computer Science

August 2023 – May 2025

University of South Florida, Tampa, Florida, United States

Certifications

- Achieved AWS Certified Solutions Architect – Associate – Amazon Web Services.
- Certified as AWS Certified Developer – Associate – Amazon Web Services.
- Earned AWS Certified Generative AI – Specialty – Amazon Web Services.
- Attained AWS Certified Cloud Practitioner – Amazon Web Services.
- Completed Microsoft Certified: Azure Fundamentals – Microsoft.
- Acquired Microsoft Power BI Data Analyst Associate – Microsoft.
- Finished SQL for Data Analysis – Udemy.